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Branching or Decision Making

Objectives



- **Topic Covered:**
 - Concept of branching
 - If-else
 - Nested if-else
 - If-else ladder

Concept of Decision Making



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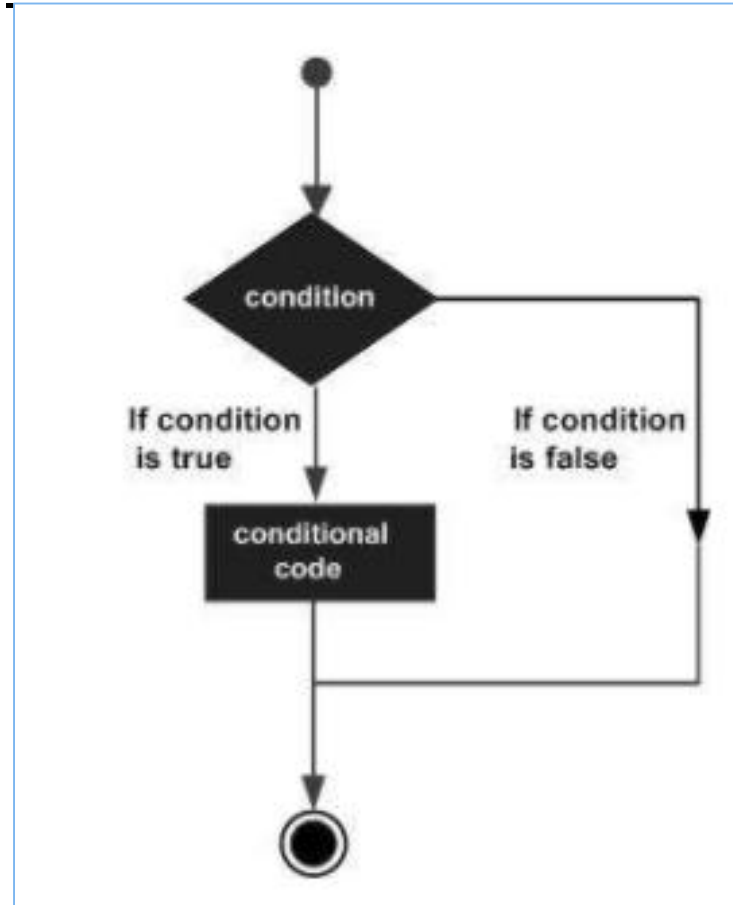
- Decision making is anticipation of conditions occurring while execution of the program and specifying actions taken according to the conditions.
- Decision structures evaluate multiple expressions which produce TRUE or FALSE as outcome. You need to determine which action to take and which statements to execute if outcome is TRUE or FALSE otherwise.

Concept of Decision Making



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- Following is the general form of a typical decision making structure found in most of the programming languages.



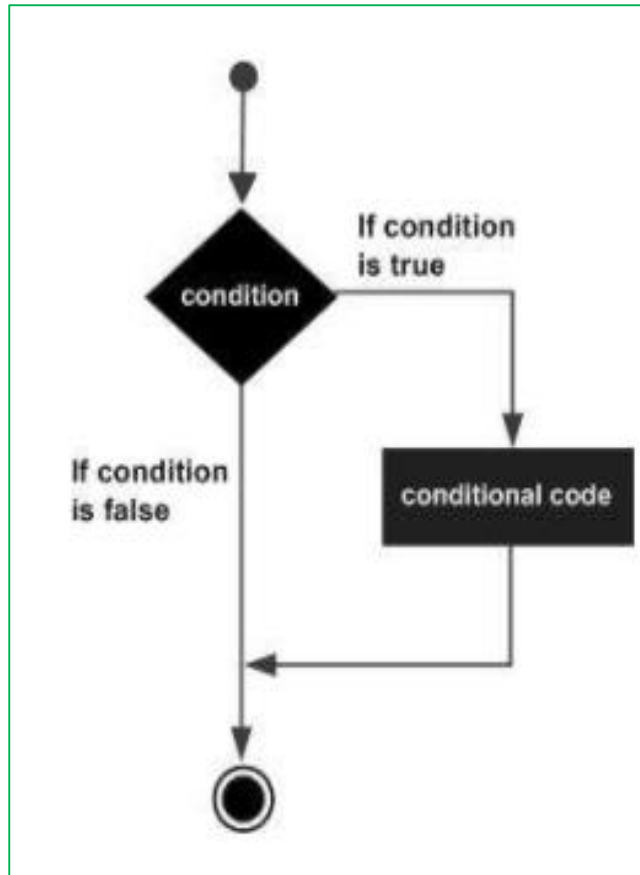
Concept of Decision Making

- Python programming language assumes any non-zero and non-null values as TRUE, and if it is either zero or null, then it is assumed as FALSE value.
- Python programming language provides following types of decision making statements. Click the following links to check their detail.

if statement

- An if statement consists of a boolean expression followed by one or more statements.
- The if statement contains a logical expression using which data is compared and a decision is made based on the result of the comparison.

```
if expression:  
    statement(s)
```



if statement

- Example

```
var1 = 100
if var1:
    print ("1 - Got a true expression value")
    print (var1)

var2 = 0
if var2:
    print ("2 - Got a true expression value")
    print (var2)
print ("Good bye!")
```

```
1 - Got a true expression value
100
Good bye!
```

Nested if statement



- There may be a situation when you want to check for another condition after a condition resolves to true. In such a situation, you can use the nested if construct.
- In a nested if construct, you can have an if...elif...else construct inside another if...elif...else construct.

Nested if statement



- Syntax

```
if expression1:
    statement(s)
    if expression2:
        statement(s)
    elif expression3:
        statement(s)
    elif expression4:
        statement(s)
    else:
        statement(s)
else:
    statement(s)
```

```
var = 100
if var < 200:
    print ("Expression value is less than 200")
    if var == 150:
        print ("Which is 150")
    elif var == 100:
        print ("Which is 100")
    elif var == 50:
        print ("Which is 50")
    elif var < 50:
        print ("Expression value is less than 50")
else:
    print ("Could not find true expression")

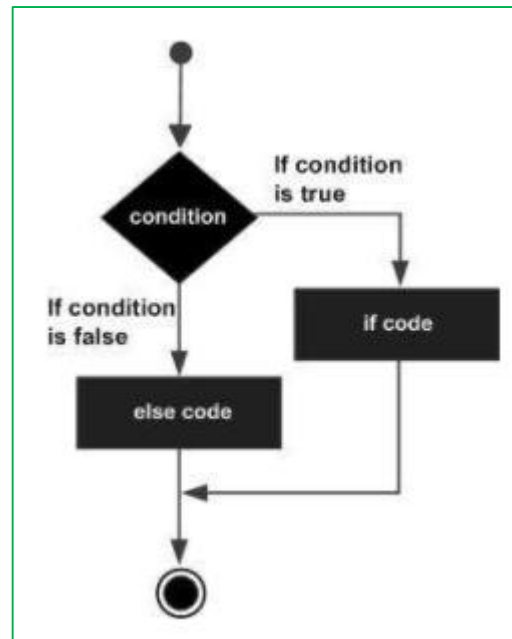
print ("Good bye!")
```

```
Expression value is less than 200
Which is 100
Good bye!
```

If-else ladder

- An else statement can be combined with an if statement. An else statement contains the block of code that executes if the conditional expression in the if statement resolves to 0 or a FALSE value.
- The else statement is an optional statement and there could be at most only one else statement following if.
- Syntax

```
if expression:  
    statement(s)  
else:  
    statement(s)
```



If-else ladder



- Example

```
var1 = 100
if var1:
    print ("1 - Got a true expression value")
    print (var1)
else:
    print ("1 - Got a false expression value")
    print (var1)

var2 = 0
if var2:
    print ("2 - Got a true expression value")
    print (var2)
else:
    print ("2 - Got a false expression value")
    print (var2)

print ("Good bye!")
```

```
1 - Got a true expression value
100
2 - Got a false expression value
0
Good bye!
```

If-else ladder



- The *elif* Statement
- The *elif* statement allows you to check multiple expressions for TRUE and execute a block of code as soon as one of the conditions evaluates to TRUE.
- Similar to the *else*, the *elif* statement is optional. However, unlike *else*, for which there can be at most one statement, there can be an arbitrary number of *elif* statements following an *if*.

If-else ladder



```
if expression1:  
    statement(s)  
elif expression2:  
    statement(s)  
elif expression3:  
    statement(s)  
else:  
    statement(s)
```

```
var = 100  
if var == 200:  
    print ("1 - Got a true expression value")  
    print (var)  
elif var == 150:  
    print ("2 - Got a true expression value")  
    print (var)  
elif var == 100:  
    print ("3 - Got a true expression value")  
    print (var)  
else:  
    print ("4 - Got a false expression value")  
    print (var)  
  
print ("Good bye!")
```

```
3 - Got a true expression value  
100  
Good bye!
```

Summary



- If-else statements can be used for decision making
- We can use nested if-else for multiple levels of checks
- If-else ladder can be used for multiple branches or multiple possibilities



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Thank You